

## Practice Problems: Calculus

ECON 441: Introduction to Mathematical Economics

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### Exercise 6.2

2. Given the function  $y = 5x^2 - 4x$ :

- (a) Find the difference quotient as a function of  $x$  and  $\Delta x$ .
- (b) Find the derivative  $dy/dx$ , using the limit definition.
- (c) Find  $f'(2)$  and  $f'(3)$ .

3. Given the function  $y = 5x - 2$ :

- (a) Find the difference quotient  $\Delta y/\Delta x$ . What type of function is it?
- (b) Since  $\Delta x$  does not appear in  $\Delta y/\Delta x$  in part (a), does it make any difference to the value of  $\Delta y/\Delta x$  whether  $\Delta x$  is large or small? Consequently, what is the limit of the difference quotient as  $\Delta x$  approaches zero?

### Exercise 7.1

3. Find  $f'(1)$  and  $f'(2)$  for the following functions:

- |                       |                                 |                         |
|-----------------------|---------------------------------|-------------------------|
| (a) $y = f(x) = 18x$  | (c) $f(x) = -5x^{-2}$           | (e) $f(w) = 6w^{1/3}$   |
| (b) $y = f(x) = cx^3$ | (d) $f(x) = \frac{3}{4}x^{4/3}$ | (f) $f(w) = -3w^{-1/6}$ |

### Exercise 7.2

3. Differentiate the following by using the product rule:

- |                      |                              |
|----------------------|------------------------------|
| (d) $(ax - b)(cx^2)$ | (e) $(2 - 3x)(1 + x)(x + 2)$ |
|----------------------|------------------------------|

7. Find the derivatives of:

- |                   |                           |
|-------------------|---------------------------|
| (a) $(x^2 + 3)/x$ | (c) $6x/(x + 5)$          |
| (b) $(x + 9)/x$   | (d) $(ax^2 + b)/(cx + d)$ |

8. Given the function  $f(x) = ax + b$ , find the derivatives of:

- (a)  $f(x)$                       (b)  $xf(x)$                       (c)  $1/f(x)$                       (d)  $f(x)/x$

### Exercise 7.3

1. Given  $y = u^3 + 2u$ , where  $u = 5 - x^2$ , find  $dy/dx$  by the chain rule.
2. Given  $w = ay^2$  and  $y = bx^2 + cx$ , find  $dw/dx$  by the chain rule.
3. Use the chain rule to find  $dy/dx$  for the following:

(a)  $y = (3x^2 - 13)^3$                       (b)  $y = (7x^3 - 5)^9$                       (c)  $y = (ax + b)^5$

4. Given  $y = (16x + 3)^{-2}$ , use the chain rule to find  $dy/dx$ . Then rewrite the function as  $y = 1/(16x + 3)^2$  and find  $dy/dx$  by the quotient rule. Are the answers identical?
5. Given  $y = 7x + 21$ , find its inverse function. Then find  $dy/dx$  and  $dx/dy$ , and verify the inverse-function rule. Also verify that the graphs of the two functions bear a mirror-image relationship to each other.
6. Are the following functions strictly monotonic? For each strictly monotonic function, find  $dx/dy$  by the inverse-function rule.

(a)  $y = -x^6 + 5 \quad (x > 0)$                       (b)  $y = 4x^5 + x^3 + 3x$